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标题	An active representation learning method for reaction yield prediction with small-scale data
作者	Hua P., Huang Z., Xu Z., Zhao Q., Ye C., Wang Y. 等 9 人
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摘要（原文节选）

Reaction optimization plays an essential role in chemical research and industrial production. To explore a large reaction system, a practical issue is how to reduce the heavy experimental load for finding the high-yield conditions. In this paper, we present an efficient machine learning tool called “ RS-Coreset ”, where the key idea is to take advantage of deep representation learning techniques to guide an interactive procedure for representing the full reaction space. Our proposed tool only uses small-scale data, s ⋯

数据表 · Table 1（8 行，含数值行 7 行）

The overlooked reaction combinations using DMF

Entry	Chloro compounds	Alkenes	LB-BH3	Solvents	Yield (%)	
Predicted	Observed					
1	a2	b6	c2	DMF	-	<5
2	a3	b3	c3	DMF	5.07	<5
3	a3	b1	c3	DMF	39.33	49
4	a1	b1	c1	DMF	59.80	43
5	a1	b2	c1	DMF	57.01	31
6	a1	b1	c3	DMF	44.68	16
7	a2	b2	c3	DMF	43.23	18

脚注（逐字）：Entry “ 1 ” is the infeasible reaction combination reported in the past46. Entry “ 2 ” - “ 7 ” are our validated reaction combinations with DMF as solvent.

正文对应句（逐字）：We first discuss how to use the existing molecular fingerprints to build the representation. In the past, several different vectorized chemical descriptors or molecular fingerprints were proposed, such as the well-known MorganFP50, Mordred51, and AvalonFP52. Recently, some fingerprints harnessing th

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